

FORESIGHT WITHOUT SEEING: LATENT FUTURES FOR WORLD ACTION MODELS

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<https://forewam.github.io/>

ABSTRACT

World Action Models (WAMs) connect visual prediction with robot control, yet supplying predictive context often comes at the cost of expensive future-video generation. Direct policies avoid this cost but lack an explicit interface for accessing future-indexed predictive information. Our key insight is that forecasting and rendering can be separated: a policy can access latent predictive computation without reconstructing future observations. Building on this insight, we introduce **ForeWAM**, a World Action Model that exposes and shapes latent predictive context for efficient control. Its core mechanism, **Future-KV**, performs a single Video DiT prefill over the current visual latent and noise-initialized future slots, then reuses the resulting key-value states throughout action denoising. Because predictive context alone does not ensure relevance to control, we further introduce dynamics registers supervised by latent actions from a frozen teacher during training, encouraging representations of interaction-induced transitions. Concentrating predictive computation in this reusable context supports a lightweight, single-layer action decoder. We evaluate ForeWAM on LIBERO, LIBERO-Plus, RoboCasa, and real-world manipulation tasks. Without additional policy-level embodied pre-training, ForeWAM improves RoboCasa success by 9.7 percentage points over Fast-WAM at the same budget of 50 demonstrations per task, reaching 59.2%. With a single-layer decoder, it achieves 77.6% success on LIBERO-Plus and reduces policy-query latency to 88.7 ms on an NVIDIA A800, delivering a $6.27\times$ speedup over Fast-WAM. These results show that exposing and shaping latent predictive computation provides useful foresight for robust, efficient control without explicit future-video generation.

1 INTRODUCTION

World Action Models (WAMs) couple visual world modeling with robot control so that action generation can benefit from predicted scene evolution (Du et al., 2023; Hu et al., 2024; Li et al., 2025; Ye et al., 2026b; Zhu et al., 2025). Yet existing WAMs face a tension between predictive context and inference efficiency. Cascaded and joint WAMs explicitly expose predicted futures to the policy, but require costly iterative video generation (Du et al., 2023; Ye et al., 2026b; Bi et al., 2026). Direct-policy WAMs such as Fast-WAM (Yuan et al., 2026b) avoid this cost by acting directly on current-observation world-model features, but consequently provide no explicit interface for future-indexed predictive computation to reach the action pathway.

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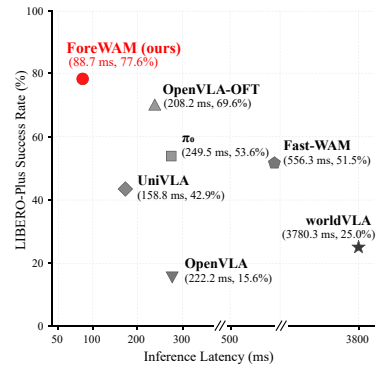


Figure 1: OOD performance vs. latency on A800.

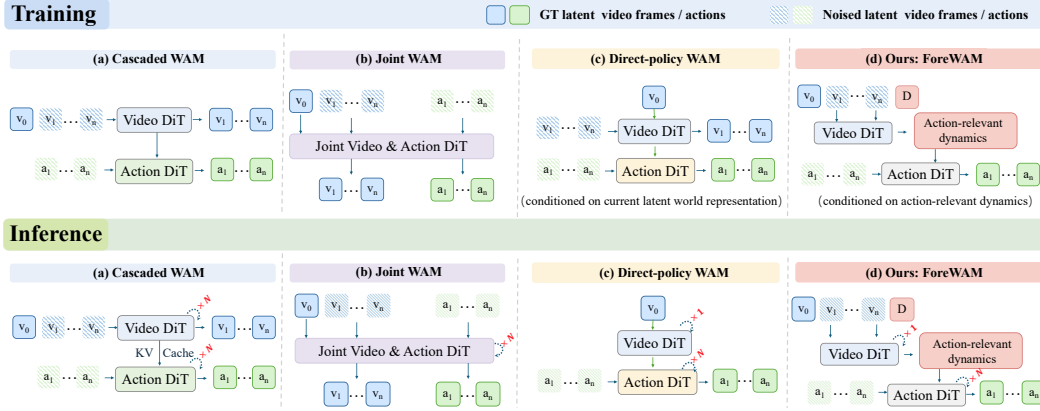


Figure 2: **World Action Model paradigms.** (a) Cascaded WAMs first generate future observations and then condition action prediction on them. (b) Joint WAMs generate future observations and actions within a unified generative process. (c) Direct-policy WAMs skip future rollout at inference and condition action prediction on a latent world representation extracted from the current observation. (d) ForeWAM retains direct action prediction while conditioning the action model on current-observation features, hidden future-slot K/V states, and dynamics registers trained with latent-action supervision. Hatched tokens denote noisy variables; future slots are stochastic internal states rather than observed future frames.

We argue that this tension stems from conflating *forecasting* with *rendering*. A future observation is only the final output of predictive computation: before a clean future frame is produced, a Video DiT already forms hidden states at future temporal positions from the current visual context, instruction, and robot state. For control, the policy need not recover the future in pixel space; it primarily needs the predictive cues encoded along the way. This suggests a different interface between world modeling and action generation: expose the latent computation of forecasting itself, rather than its rendered output. In other words, a policy may obtain *foresight without seeing the future*.

Based on this insight, we introduce **ForeWAM**, a **Foresight-without-Seeing World Action Model**. Its core mechanism, **Future-KV**, exposes future-indexed predictive states without generating future video. ForeWAM appends noise-initialized future slots to the current visual latent and performs a single Video DiT prefill. The resulting key-value states from the current and future positions are cached and reused throughout action denoising. The action policy can therefore access predictive context computed by the video model, while avoiding iterative future-video denoising and decoding.

Yet predictive context is not necessarily *action-relevant* context. Video prediction models broad scene evolution, including many changes that may be irrelevant to manipulation, whereas action generation depends particularly on interaction-induced transitions and task progress. We therefore introduce **dynamics registers** and supervise them with latent actions extracted by a frozen LaWM teacher (Chen et al., 2026a) from demonstrated current-future observation pairs. This training-only signal shapes the exposed predictive representation toward dynamics that matter for control. The distinction is strongly reflected in our ablation: on LIBERO-Plus, Future-KV alone improves the matched baseline from 53.6% to 58.5%, and latent-action supervision alone reaches 54.8%, whereas combining them yields 76.7% (Table 1).

With predictive reasoning concentrated in the reusable video-side context, the action branch can be substantially simplified. The Video DiT is evaluated only once per policy query, whereas the Action DiT runs repeatedly throughout action denoising; shifting computation toward the former is therefore particularly beneficial for inference. We find that a single-layer Action DiT reaches 77.6% on LIBERO-Plus, compared with 76.7% for a 30-layer decoder, while a one-layer Fast-WAM reaches only 57.9% (Table 2). This suggests that the shallow decoder is enabled by the richer predictive context, rather than action decoding being intrinsically easy.

Experiments across LIBERO, LIBERO-Plus, RoboCasa, and real-world manipulation validate this design. Without additional policy-level embodied pretraining, ForeWAM achieves 59.2% success on RoboCasa with only 50 demonstrations per task, outperforming Fast-WAM by 9.7 percentage points under the same demonstration budget. It retains near-saturated performance on standard LIBERO

while substantially improving robustness under the distribution shifts of LIBERO-Plus. Meanwhile, ForeWAM reduces policy-query latency from 556.3 ms to 88.7 ms on an NVIDIA A800, a $6.27\times$ speedup over Fast-WAM (Figure 1). Together, these results show that useful world-model foresight need not be materialized as future video: it can instead be exposed and shaped directly in latent space for efficient control.

2 RELATED WORK

Vision-language-action policies. VLA models map visual observations and language instructions to executable robot actions (Brohan et al., 2022; 2023; Kim et al., 2024; Liu et al., 2025; Team et al., 2024; Yang et al., 2026), with many recent models adapting pretrained vision-language backbones for control (Black et al., 2024; Intelligence et al., 2025; Zhao et al., 2025). Diffusion and flow objectives support multimodal continuous action generation (Black et al., 2024; Chi et al., 2025; Lipman et al., 2022), while large-scale robot pretraining can improve transfer across tasks and embodiments (Bjorck et al., 2025; Bu et al., 2025; Zheng et al., 2026). However, the action-prediction objective alone does not explicitly require learning predictive representations of how the environment evolves during interaction, motivating the integration of world modeling into policy learning.

World-action models. World Action Models (WAMs) couple world prediction with action generation through cascaded or joint architectures. Cascaded approaches follow an *imagine-then-act* pipeline, extracting actions from generated future visual observations as intermediate plans (Du et al., 2023; 2024), geometric representations (Bharadhwaj et al., 2024; Ko et al., 2024; Xu et al., 2024; Zhi et al., 2025) or predictive latent features (Hu et al., 2024; Huang et al., 2024; Lou et al., 2026; Yan et al., 2026b). Joint WAMs learn future states and actions together. Autoregressive variants organize them within a unified temporal sequence (Cen et al., 2025a;b; Cheang et al., 2024; Wu et al., 2024; Team et al., 2026). Diffusion- and flow-based variants use either unified prediction backbones (Chen et al., 2026b; Guo et al., 2024; Kim et al., 2026; Shen et al., 2026; Ye et al., 2026b; Zhao et al., 2026; Zheng et al., 2025; Zhu et al., 2025) or coordinated modality-specific streams, coupling their predictive streams through shared attention (Bi et al., 2026; Lyu et al., 2026; Won et al., 2025; Yang et al., 2025; Yuan et al., 2026a), intermediate hidden states (Li et al., 2026a; Ma et al., 2026), or shared representations (Li et al., 2025). Iterative future generation and tightly coupled world-action computation can incur substantial inference overhead. Direct-policy WAMs, including Fast-WAM and GigaWorld-Policy, retain world-modeling supervision during training but use action-only inference (Ye et al., 2026a; Yuan et al., 2026b). Concurrent work explores complementary forms of predictive supervision: Enfold (Zeng et al., 2026) distills world-generator states into representations predicted from current observations and language, while Robust-WAM (Yan et al., 2026a) aligns action-stream queries with future visual semantics. ForeWAM retains noisy future slots at inference and conditions action denoising on their hidden K/V states from a single Video DiT prefill, together with dynamics registers supervised by LaWM latent-action targets (Chen et al., 2026a), without generating future video.

3 FROM PREDICTIVE CONTEXT TO ACTIONABLE FORESIGHT

To understand what an action policy actually needs from a video world model, we conduct a controlled study on LIBERO-Plus using Fast-WAM retrained with Wan2.1 as the matched direct-policy baseline (Yuan et al., 2026b). We focus on two questions: whether future-indexed latent context is sufficient for control, and how the resulting context changes the capacity required by the action pathway. The observations below motivate the design of ForeWAM in Section 4.

3.1 USEFUL FORESIGHT REQUIRES BOTH ACCESS AND SHAPING

We first examine how predictive information should be exposed to the action policy. The matched Fast-WAM baseline conditions action generation on current-observation world-model features and achieves 53.6% average success. Providing additional future-indexed context through Future-KV improves this result to 58.5% (Table 1), despite requiring no explicitly generated future observation at inference. This suggests that internal states associated with future temporal positions provide useful context beyond current-observation features alone.

Table 1: **Predictive-context design study on LIBERO-Plus (%)**. Future-KV exposes future-indexed context, while LA supervision shapes interaction-relevant dynamics. All configurations use a 30-layer Action DiT under matched settings. Bold marks the best result in each column.

LA	Future-KV	Camera	Robot	Language	Light	Background	Noise	Layout	Avg
×	×	37.0	36.5	64.1	81.5	49.3	50.1	63.7	53.6
✓	×	49.0	36.8	65.1	75.9	48.5	50.8	61.7	54.8
×	✓	37.8	41.0	63.9	83.4	65.7	61.5	65.6	58.5
✓	✓	64.6	61.6	86.6	95.3	74.1	77.7	81.7	76.7

However, access to such context is clearly not sufficient. Video prediction models scene evolution broadly, whereas control depends particularly on interaction-induced transitions and task progress. Consistent with this distinction, latent-action (LA) supervision alone provides only a marginal gain, from 53.6% to 54.8%, while Future-KV alone reaches 58.5%. Strikingly, combining the two increases success to 76.7%, outperforming the stronger individual component by 18.2 percentage points. The same combination improves all seven LIBERO-Plus perturbation categories over either component alone (Table 9 in Appendix D).

These results reveal two complementary requirements for useful foresight. Future-KV determines *what predictive context is accessible* to the action pathway, whereas LA supervision shapes *what information that context emphasizes*. In other words, predictive context becomes substantially more useful for control when future-indexed representations are both **exposed** and **shaped toward interaction-relevant dynamics**.

3.2 RICH PREDICTIVE CONTEXT ENABLES LIGHTWEIGHT ACTION DECODING

The representation supplied by the video model also determines how much computation remains necessary in the action pathway. The Video DiT constructs context once per policy query, whereas the Action DiT is evaluated repeatedly throughout action denoising. If the video-side context already captures the dynamics needed for control, a deep action decoder may therefore be redundant.

Table 2: Action-expert depth comparison. LIBERO-Plus success rate (%), without additional policy-level embodied pretraining. All configurations retain the 30-layer Wan2.1 Video DiT; action-layer mappings and initializations follow Appendix A. Bold marks the best result in each column.

Action DiT depth	Camera	Robot	Language	Light	Background	Noise	Layout	Avg
Fast-WAM / Wan2.1 (30 layers)	37.0	36.5	64.1	81.5	49.3	50.1	63.7	53.6
Fast-WAM / Wan2.1 (1 layer)	44.9	38.0	71.5	81.6	51.3	54.2	69.0	57.9
ForeWAM (30 layers)	64.6	61.6	86.6	95.3	74.1	77.7	81.7	76.7
ForeWAM (12 layers)	67.7	63.7	86.0	90.8	71.8	80.0	80.5	76.9
ForeWAM (6 layers)	61.5	61.4	79.6	92.2	76.4	78.6	78.4	74.7
ForeWAM (1 layer)	61.7	56.8	85.1	95.8	76.5	89.1	83.0	77.6

We test this hypothesis by retaining all 30 layers of the Video DiT while varying the Action DiT depth. ForeWAM achieves 76.7%, 76.9%, 74.7%, and 77.6% average success with 30, 12, 6, and 1 action layers, respectively (Table 2). Increasing action depth provides no monotonic benefit; remarkably, the single-layer decoder achieves the highest overall success among these variants.

This result is not simply explained by action decoding being intrinsically easy. Reducing the matched Fast-WAM action branch from 30 layers to one improves its success only from 53.6% to 57.9%. At the same one-layer depth, ForeWAM reaches 77.6%, a 19.7-point advantage. The lightweight decoder is therefore effective in conjunction with the richer predictive context, rather than because the original action pathway is unnecessarily deep.

Together, these observations suggest a clear division of computation: the video branch should expose and organize predictive dynamics into an action-relevant representation, while the repeatedly evaluated action branch can remain lightweight. ForeWAM instantiates this principle with Future-KV, latent-action-supervised dynamics registers, and a shallow Action DiT, which we detail next.

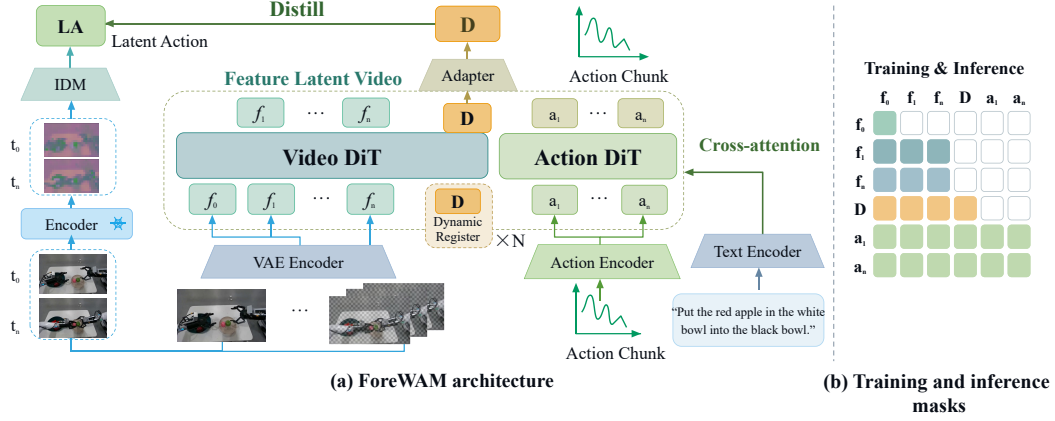


Figure 3: **ForeWAM**. (a) Architecture and training pipeline. A supervision pass uses demonstrated futures for video and latent-action learning, while an independent action-conditioning pass uses pure-noise future slots to construct the predictive context consumed by the action policy. At inference, only the latter is required: one Video DiT prefill supplies cached K/V throughout action denoising, without future observations or the latent-action teacher. (b) Structured attention mask shared by training and inference. Rows are queries and columns are keys; colored cells indicate allowed attention. Here $f_0 = C$, $f_{1:n} = F$, and $a_{1:n} = A$. Future slots attend bidirectionally to one another.

4 FOREWAM

Guided by the observations in Section 3, we introduce **ForeWAM**, a direct-policy World Action Model that exposes predictive context from a video world model without explicitly generating future video. ForeWAM follows a simple decomposition. **Future-KV** extracts future-indexed latent context through a single Video DiT prefill; learnable **dynamics registers** shape this context toward interaction-relevant transitions through latent-action supervision; and a lightweight Action DiT reuses the resulting predictive context throughout action denoising. Figure 3 illustrates the overall architecture.

Problem formulation. Given synchronized multi-camera observations o , a language instruction l , and proprioceptive state p , the policy predicts an executable action chunk $a_{1:H} \in \mathbb{R}^{H \times d_{\text{act}}}$. We retain the direct-policy interface $p_\theta(a_{1:H} \mid o, l, p)$ (Yuan et al., 2026b). Let z_{cur} denote the current-frame latent encoded by the video VAE, and $z_{1:T}$ the demonstrated video latents used for training supervision. Future observations and latent-action teacher outputs are used only during training.

4.1 LATENT FORESIGHT THROUGH FUTURE-KV

The first component exposes predictive computation associated with future temporal positions without materializing future observations. For the action-conditioning pathway, we concatenate the clean current latent with Gaussian future slots:

$$z^{\text{init}} = [z_{\text{cur}}; \epsilon_F], \quad \epsilon_F \sim \mathcal{N}(0, I), \quad (1)$$

where $[\cdot; \cdot]$ denotes temporal concatenation. The Video DiT processes this sequence once at $\sigma = 1.0$, conditioned on the instruction and proprioception. We retain the resulting hidden key-value states and reuse them throughout action denoising. We refer to this interface as **Future-KV**.

Importantly, the future slots contain no observed future information. Instead, they provide future temporal positions at which a video model trained with future supervision can construct contextual states from the available inputs. ForeWAM exposes these internal states directly to the action pathway rather than iteratively denoising them into valid future frames. Thus, predictive context is computed once and reused, while explicit future-video generation is avoided.

4.2 SHAPING PREDICTIVE CONTEXT WITH LATENT ACTIONS

Future-KV provides access to future-indexed context, but does not by itself specify which aspects of scene evolution should be emphasized for control. We therefore introduce learnable dynamics registers $D = \{D_i\}_{i=1}^{N_D}$ into the Video DiT and supervise their pooled representation with a latent-action target extracted from demonstrated transitions.

Latent-action supervision. We use a frozen LaWAM teacher (Chen et al., 2026a) to encode a demonstrated transition. Let o_{ext} and o_{ext}^+ denote the current and future observations from the external-camera view. A frozen visual encoder f_{vis} and latent-action extraction mapping E_{IDM} produce

$$\begin{aligned} u_{\text{cur}} &= f_{\text{vis}}(o_{\text{ext}}), & u_{\text{fut}} &= f_{\text{vis}}(o_{\text{ext}}^+), \\ z_{\text{LA}} &= E_{\text{IDM}}(u_{\text{cur}}, u_{\text{fut}}). \end{aligned} \quad (2)$$

The teacher representation z_{LA} describes the demonstrated transition and is used only as a training target. It is distinct from the video VAE latents $z_{1:T}$ and is never supplied directly to the action decoder. A trainable projection g_ψ aligns the pooled dynamics-register representation with this latent-action target.

Routing predictive context. ForeWAM organizes information flow among current-frame tokens C , future-slot tokens F , dynamics registers D , and action tokens A using the structured attention mask in Figure 3(b). Within the video branch, current-frame tokens attend only to C , while future-slot tokens attend to C and all tokens in F , with bidirectional attention across future slots. Dynamics registers attend to C , F , and D , allowing them to aggregate current and future-indexed information without feeding back into the visual tokens.

The resulting video-side context is exposed to the action branch through cached K/V. At each action-denoising step, action queries attend to cached states from C , F , and D , together with the current action-token K/V from A . Both branches additionally receive text embeddings and projected proprioceptive conditioning, which are omitted from the mask diagram for clarity.

4.3 LIGHTWEIGHT ACTION DECODING FROM CACHED PREDICTIVE CONTEXT

As shown in Section 3.2, once the video branch provides rich predictive context, the action pathway can be substantially simplified. We therefore use a single-layer Action DiT that reads the final Video DiT layer, $L = 30$.

A single video prefill produces the dynamics-register states and final-layer K/V cache:

$$(D, K, V) = \text{Prefill}_\phi(z^{\text{init}}, l, p), \quad (3)$$

where $K = K^{(L)}$ and $V = V^{(L)}$ contain the final-layer states of the current-frame, future-slot, and dynamics-register tokens. These features condition action prediction as

$$p_\theta(a_{1:H} \mid o, l, p, D, K, V). \quad (4)$$

The dependence of (D, K, V) on (o, l, p, ϵ_F) is implicit.

At inference, the Video DiT is evaluated only once for each action-chunk prediction. During subsequent action denoising, the single-layer Action DiT repeatedly reads the same cached predictive context together with the evolving action tokens. ForeWAM therefore concentrates visual and predictive computation in a reusable video-side representation while keeping repeated action-side computation lightweight.

4.4 TRAINING AND INFERENCE

Training uses two Video DiT forward passes with shared parameters but independent hidden states and K/V caches. The *supervision pass* receives noised demonstrated futures and provides the video-prediction and latent-action objectives. The independent *action-conditioning pass* instead receives freshly sampled pure-noise future slots at $\sigma = 1.0$, matching the inference-time input in Eq. 1, and produces (D, K, V) for the action objective. This separation ensures that demonstrated future observations supervise the model during training without being directly exposed to the action-conditioning pathway.

Video flow matching. The video branch learns demonstrated visual dynamics through continuous flow matching (Lipman et al., 2022). Writing the target video latents as $z = z_{1:T}$, we sample Gaussian noise ϵ_z and flow time $\tau \in (0, 1)$ to construct $z_\tau = (1 - \tau)z + \tau\epsilon_z$, while keeping the current-frame latent clean. The video velocity predictor v_ϕ is optimized with

$$\mathcal{L}_{\text{video}} = \mathbb{E}_{z, \epsilon_z, \tau} \left[\|v_\phi(z_\tau, \tau \mid o, l, p) - (\epsilon_z - z)\|_2^2 \right]. \quad (5)$$

Let D^{sup} denote the dynamics-register states produced by this supervision pass.

Action flow matching. Let $a = a_{1:H}$ be the demonstrated action chunk. We sample Gaussian noise ϵ_a and action flow time $s \in (0, 1)$, and form $a_s = (1 - s)a + s\epsilon_a$. Conditioned on the predictive context (D, K, V) from the independent pure-noise prefill, the Action DiT predicts the action velocity:

$$\mathcal{L}_{\text{action}} = \mathbb{E} \left[\|v_\theta(a_s, s \mid o, l, p, D, K, V) - (\epsilon_a - a)\|_2^2 \right]. \quad (6)$$

The expectation is over training examples, action noise and flow time, and fresh future-slot noise ϵ_F . Crucially, the noised demonstrated future z_τ , the supervision-pass registers D^{sup} , and their K/V states never enter this action-conditioning forward pass.

Latent-action supervision. The projection g_ψ aligns the mean-pooled supervision-pass dynamics registers with the frozen latent-action target from Eq. 2:

$$\mathcal{L}_{\text{LA}} = \left\| g_\psi \left(\frac{1}{N_D} \sum_{i=1}^{N_D} D_i^{\text{sup}} \right) - \text{sg}(z_{\text{LA}}) \right\|_2^2, \quad (7)$$

where sg stops gradients through the frozen teacher. The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{video}} + \mathcal{L}_{\text{action}} + \lambda_{\text{LA}} \mathcal{L}_{\text{LA}}. \quad (8)$$

Action gradients propagate through the video-to-action K/V interface, allowing the predictive context to be optimized jointly for control.

Inference. At inference, the supervision pass, demonstrated future observations, and latent-action teacher are all removed. ForeWAM encodes the current observation, samples pure-noise future slots, performs a single Video DiT prefill, and reuses the resulting (D, K, V) cache throughout action denoising. No future video is generated or decoded.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

We evaluate ForeWAM on LIBERO, RoboCasa, the OOD LIBERO-Plus benchmark, and real-world manipulation tasks, using success rate as the primary metric.

Benchmarks and protocols. We train on the original LIBERO demonstrations (500 per suite) and evaluate on its four standard suites and the seven LIBERO-Plus perturbation categories, without using augmented LIBERO-Plus training data (Liu et al., 2023; Fei et al., 2025). We evaluate RoboCasa on 24 static kitchen tasks over three seeds (3,600 trials), using successful human demonstrations without additional MimicGen data.

Real-world experiments. We jointly train on 125 teleoperated demonstrations (71,702 frames) from four tabletop tasks using a 6-DoF AgileX Piper arm with third-person and wrist RGB cameras. ID evaluation uses 10 trials per task with matched randomized object placements; OOD evaluation perturbs object attributes in pen placement and drawer manipulation (Figure 4). Full benchmark protocols and task definitions appear in Appendix F; additional real-world details are in Appendix C.

5.2 MAIN RESULTS

Results on RoboCasa. ForeWAM achieves strong manipulation performance with a limited demonstration budget (Table 3). With 50 demonstrations per task and no additional policy-level

Table 3: Results on RoboCasa.

Method	P.T.	Demos/ task	SR (%)
GR00T-N1 (Bjorck et al., 2025)	✓	300	49.6
DP-VLA (Han et al., 2024)	✓	3000	57.3
GR00T-N1 + DreamGen (Jang et al., 2025)	✓	300	57.6
GR00T-N1 + DUST (Won et al., 2025)	✓	300	58.5
UVA (Li et al., 2025)	×	50	50.0
Fast-WAM (Yuan et al., 2026b)	×	50	49.5
ForeWAM	×	50	59.2

Table 4: Results on LIBERO.

Method	P.T.	Avg.
OpenVLA (Kim et al., 2024)	✓	76.5
WorldVLA (Cen et al., 2025b)	✓	81.8
π_0 -Fast (Pertsch et al., 2025)	✓	85.5
π_0 (Black et al., 2024)	✓	94.1
UniVLA (Bu et al., 2025)	✓	95.2
$\pi_{0.5}$ (Intelligence et al., 2025)	✓	96.9
CogVLA (Li et al., 2026b)	✓	97.4
Fast-WAM (Yuan et al., 2026b)	×	97.6
ForeWAM	×	97.9

P.T.: policy-level embodied pretraining (Appendix A).

embodied pretraining, it reaches 59.2% average success across 24 kitchen manipulation tasks, compared with 49.5% for Fast-WAM under the same demonstration budget and pretraining setting, a gain of 9.7 percentage points. ForeWAM also exceeds GR00T-N1 (Bjorck et al., 2025) by 9.6 percentage points while using one-sixth as many demonstrations per task (50 versus 300), achieving the highest success rate among the methods listed in the table.

Results on LIBERO & LIBERO-Plus. ForeWAM preserves the near-saturated performance of Fast-WAM on standard LIBERO while substantially improving robustness under the distribution shifts evaluated by LIBERO-Plus (Tables 4 and 5). On LIBERO, ForeWAM achieves 97.9% average success compared with 97.6% for Fast-WAM, with neither method using additional policy-level embodied pretraining. Suite-level results are provided in Appendix E. On the full LIBERO-Plus benchmark, the improvement is much larger: ForeWAM raises average success from 51.5% to 77.6%, a gain of 26.1 percentage points, and exceeds OpenVLA-OFT by 8.0 points. It leads five of the seven perturbation categories, including lighting (95.8%) and sensor noise (89.1%). These results show that the main gain over Fast-WAM lies in robustness to the evaluated distribution shifts, while standard manipulation performance remains high.

Table 5: Success rates (%) on seven LIBERO-Plus perturbation categories.

Method	Params	P.T.	Camera	Robot	Language	Light	Background	Noise	Layout	Avg
OpenVLA (Kim et al., 2024)	7B	✓	0.8	3.5	23.0	8.1	34.8	15.2	28.5	15.6
OpenVLA-OFT	7.5B	✓	56.4	31.9	<u>79.5</u>	<u>88.7</u>	93.3	75.8	<u>74.2</u>	<u>69.6</u>
π_0 (Black et al., 2024)	3.3B	✓	13.8	6.0	58.8	85.0	<u>81.4</u>	<u>79.0</u>	68.9	53.6
π_0 -Fast (Pertsch et al., 2025)	3.3B	✓	65.1	21.6	61.0	73.2	<u>73.2</u>	<u>74.4</u>	68.8	61.6
UniVLA (Bu et al., 2025)	7B	✓	1.8	<u>46.2</u>	69.6	69.0	81.0	21.2	31.9	42.9
WorldVLA (Cen et al., 2025b)	7B	✓	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
Fast-WAM (Yuan et al., 2026b)	6B	×	16.4	44.5	68.9	78.2	53.7	37.7	60.7	51.5
ForeWAM	1.5B	×	<u>61.7</u>	56.8	85.1	95.8	76.5	89.1	83.0	77.6

Results on Real-World Tasks. ForeWAM achieves the highest average success rate across the four in-distribution (ID) real-world tasks (Figure 4). Without additional policy-level embodied pretraining, it reaches 92.5% success, outperforming Fast-WAM by 42.5 percentage points and π_0 by 5.0 points. The largest gain over Fast-WAM occurs on placing blocks in the drawer and closing it, where success increases from 0% to 90%. Under object-attribute perturbations in the two OOD tasks, ForeWAM achieves 40.0% average success, compared with 10.0% for Fast-WAM and 15.0% for Cosmos Policy (Kim et al., 2026). While ForeWAM outperforms both π_0 and $\pi_{0.5}$ on ID tasks, all three evaluated WAMs remain below π_0 (60.0%) and $\pi_{0.5}$ (45.0%) in average OOD success. This shared gap may partly reflect differences in pretraining, as ForeWAM, Fast-WAM, and Cosmos Policy do not undergo the large-scale policy-level robot pretraining used by these two policies.

5.3 ANALYSIS

Action-to-image Attention. Figure 5 visualizes the action-to-image attention maps of Fast-WAM and ForeWAM across two manipulation examples. ForeWAM exhibits more localized attention

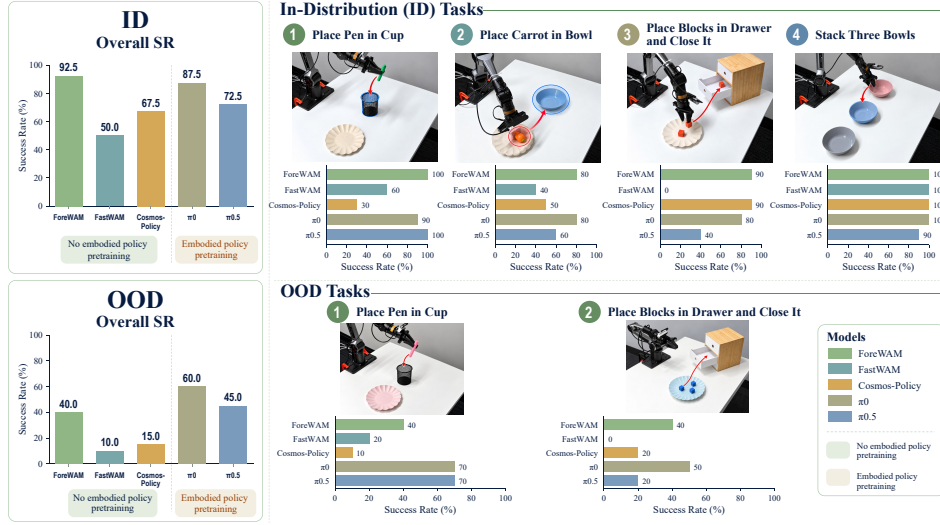


Figure 4: Real-world task success rates (%). Top: four in-distribution tasks. Bottom: OOD evaluation with perturbed object attributes in pen placement and drawer manipulation. Left panels report the mean success rate across the tasks in each setting; right panels show task examples and per-task results. ForeWAM and Fast-WAM use no additional policy-level embodied pretraining.

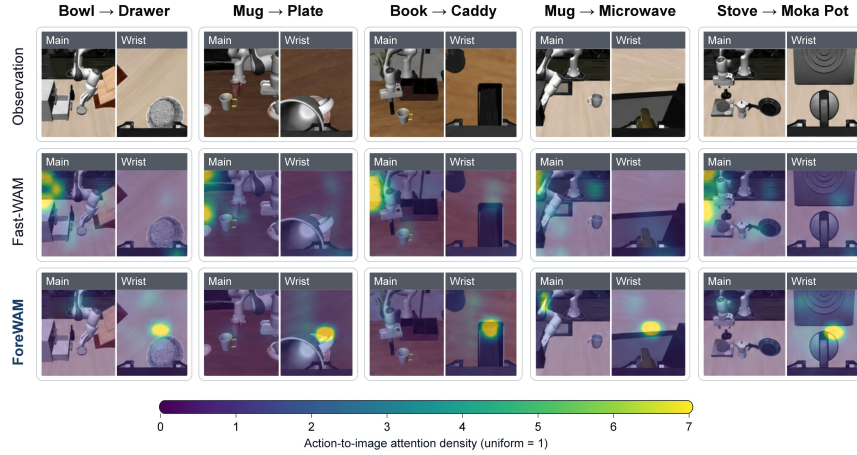


Figure 5: Attention maps.

around interaction-relevant regions, including manipulated objects, target receptacles, and contact areas, particularly in wrist views, while Fast-WAM shows more diffuse responses and stronger activations in less task-relevant regions. This indicates that ForeWAM provides more interaction-focused visual context to the action pathway.

Inference Efficiency. Table 6 compares overall and component-wise inference latency on LIBERO-10 using a single NVIDIA A800 GPU and 10 action-denoising steps. ForeWAM reduces overall latency from 556.3 ms to 88.7 ms, yielding a $6.27\times$ speedup over Fast-WAM.

The largest reduction occurs in action generation, which decreases from 470.6 ms to 21.7 ms, while visual processing remains comparable (46.4 ms versus 47.7 ms). Thus, the measured speedup primarily comes from the action-generation stage. ForeWAM’s single Video DiT prefill supplies cached Future-KV features to the one-layer Action DiT throughout denoising, retaining latent future condi-

Table 6: Inference latency (ms) on LIBERO-10.

Stage	Fast-WAM	ForeWAM
VAE Enc.	12.7	10.0
Visual	47.7	46.4
Action	470.6	21.7
Other	25.3	10.6
Overall	556.3	88.7

tioning with a lightweight action pathway. Together with the success-rate results above, these measurements show that ForeWAM combines strong manipulation performance with low inference latency. Appendix B provides details of the latency comparison.

6 LIMITATIONS AND DISCUSSION

Our evaluation covers LIBERO, LIBERO-Plus, RoboCasa, and real-world tabletop manipulation. However, the real-world experiments are limited to a single robot platform, four tasks, and a small number of trials. Generalization under the evaluated object-attribute shifts remains challenging, while transfer to other embodiments and broader task distributions has yet to be established. Extending the evaluation to more diverse robot platforms, interaction dynamics, and longer-horizon tasks is an important direction for future work.

7 CONCLUSION

We introduced ForeWAM, a direct-policy World Action Model that provides predictive context for control without explicit future-video generation. ForeWAM combines Future-KV with latent-action-supervised dynamics registers: a single Video DiT prefill supplies cached context that a lightweight Action DiT reuses throughout action denoising. Future observations and the latent-action teacher are required only during training.

Experiments across simulation benchmarks and real-world manipulation tasks demonstrate the effectiveness of this design, while component ablations show the complementary benefits of Future-KV and latent-action supervision. With a single-layer action decoder, ForeWAM achieves 77.6% success on LIBERO-Plus and a policy-query latency of 88.7 ms on an NVIDIA A800. These results support exposing and shaping latent predictive context as an effective approach to robust, efficient control without rendering future observations.

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A TRAINING DETAILS

Architecture and inputs. We initialize the visual branch from Wan2.1-T2V-1.3B, retaining its video DiT, text encoder, and video VAE (Wan et al., 2025). We precompute instruction embeddings with the corresponding Wan2.1 text encoder. The video DiT comprises 30 transformer blocks, while the default action expert uses one block and attends to the final-layer video K/V cache. We additionally evaluate action experts with 6, 12, and 30 blocks in the depth ablation; the component and register-teacher ablations use 30-block action experts. The video branch uses hidden dimension $d_v = 1536$, whereas the action expert uses $d_a = 1024$ and is initialized from a linearly interpolated Wan2.1 ActionDiT checkpoint. The action horizon is $H = 32$.

Each training example spans 33 observation time steps and a 32-step action chunk. The data loader samples video frames at indices $0, 4, \dots, 32$, yielding nine RGB frames. The VAE then compresses these to $1 + (9 - 1)/4 = 3$ latent temporal positions: one current-frame position and two future slots. The teacher observation pair uses the first and last sampled frames, corresponding to original indices 0 and 32. For LIBERO, we concatenate the two synchronized camera views along the image width before VAE encoding, producing a 224×448 image composed of two 224×224 views. The policy additionally receives an 8-dimensional proprioceptive state. Each action is seven-dimensional, comprising a 6-DoF end-effector pose and one gripper-control dimension.

Optimization and inference. We jointly optimize the video expert, action expert, dynamics registers, latent-action projection, and proprioceptive encoder using AdamW, with the settings in Table 7. Video and action training use continuous flow matching. Latent-action supervision uses mean-squared error against detached targets from the frozen LaWM teacher. We disable readability registers and allow action gradients to propagate through the video K/V interface. At inference, one video prefill at $\sigma = 1.0$ supplies the cache reused throughout action denoising. None of the reported variants receives additional policy-level embodied pretraining before LIBERO training.

Table 7: Shared training and inference hyperparameters.

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.01
Learning-rate schedule	Cosine decay
Maximum gradient norm	1.0
Flow schedules	1,000 timesteps; shift 5.0
Video / action / latent-action loss weights	1.0 / 1.0 / 1.0
Dynamics registers / teacher target dimension	16 / 32
Action horizon / inference denoising steps	32 / 10
Training seed	42

Training budgets. Table 8 summarizes the dataset-specific training configurations. Piper uses BF16 with an online FP32 LaWM teacher and produces a checkpoint at 10,500 optimizer updates.

Table 8: Dataset-specific training configurations.

Setting	LIBERO	RoboCasa	Piper
Epochs	20	20	20
Global batch size	128	128	128

Action-to-video layer mapping. Each action block reads K/V from a specified video block. With zero-based video block indices, the mappings for action depths 1, 6, and 12 are [29], [0, 6, 12, 17, 23, 29], and [0, 3, 5, 8, 11, 13, 16, 18, 21, 24, 26, 29], respectively. The 30-block action expert uses the identity mapping [0, 1, . . . , 29]. Each depth uses its corresponding interpolated Action DiT initialization. The video expert retains all 30 blocks in every case.

Policy initialization and pretraining scope. The Video DiT is initialized from pretrained Wan2.1 weights, and the Action DiT backbone is initialized from the same checkpoint through layer mapping and weight interpolation. We do not perform an additional large-scale robot-data pretraining stage for either branch before task-specific training.

Separate training forward passes. For Future-KV with Video DiT registers, training requires two video forward passes to prevent demonstrated future observations from leaking into the action-conditioning features, matching the information available at inference. The first combines the clean current latent with noised demonstrated future latents and supplies the video loss and register states for LaWM supervision. The second combines the clean current latent with newly sampled Gaussian future slots at $\sigma = 1.0$ and recomputes the register states and layer-wise K/V, including register-token K/V, for the action loss. The default one-layer Action DiT reads the final-layer cache. These passes share model parameters and learnable register initializations, but do not reuse each other’s hidden states or caches. Demonstrated futures influence the action policy through parameter learning, not as inputs to its conditioning pass. Only the second video pass is required at inference.

Frozen LaWM teacher. We use the publicly released LaWM checkpoint from LaWAM (Chen et al., 2026a) as a frozen teacher for latent-action supervision. The teacher provides training targets and is discarded at inference. Its pretraining on external embodied data provides indirect access to external embodied knowledge during policy training; this data provenance remains relevant to comparisons even though the teacher is absent at inference.

P.T. reporting convention. In the tables, P.T. denotes large-scale embodied-data pretraining of the policy backbone or action expert before task-specific training, including inherited weights from an embodied-pretrained policy. It excludes general-purpose video pretraining and pretraining of auxiliary frozen teachers. Under this convention, ForeWAM is marked as not using policy-level embodied pretraining, while its use of the pretrained LaWM teacher is disclosed above. This marking does not imply that the complete training pipeline is free of external embodied knowledge.

Ablation baseline. The baseline with both components disabled in Table 1 is Fast-WAM retrained with Wan2.1 under the same experimental configuration as the 30-layer ForeWAM ablations. It achieves 53.6% overall success on LIBERO-Plus and is distinct from the 6B Fast-WAM configuration reported in Table 5 (51.5%).

B DETAILS OF THE LATENCY COMPARISON

Evaluation setting. The comparison in Table 6 measures the time required for one policy query on LIBERO-10 using a single NVIDIA A800 GPU with 80 GB of memory. Both methods use 10 action-denoising steps. We perform 5 warm-up iterations before collecting 10 timed runs and report the arithmetic mean in milliseconds. We time action-chunk prediction, excluding environment stepping, rendering, and episode resets.

ForeWAM configuration. ForeWAM uses the Wan2.1-T2V-1.3B visual branch and the one-layer Action DiT described in Appendix A. The two camera views form a 224×448 input, accompanied

by the current proprioceptive state. The model encodes the current observation and performs one Video DiT prefill over the current latent, noisy future slots, and dynamics registers. The action expert reuses the final-layer video K/V cache across all 10 denoising steps to produce a 32-action chunk. Readability registers are disabled. Neither future observations nor the latent-action teacher is required at inference, and no future video is decoded.

Timing breakdown. *VAE Enc.* measures encoding of the current visual observation into latent space. *Visual* covers visual-context computation, including ForeWAM’s Video DiT prefill and construction of the K/V cache. *Action* covers action generation across the full denoising loop. *Other* accounts for the remaining inference overhead outside these three stages, while *Overall* reports the total policy-query latency. For the reported means, these components sum to 88.7 ms for ForeWAM and 556.3 ms for Fast-WAM.

Latency reduction. ForeWAM reduces total latency by 84.1%, corresponding to a $6.27\times$ speedup. Action generation accounts for 448.9 ms of the 467.6 ms total saving, whereas visual-context computation differs by only 1.3 ms.

C REAL-WORLD EXPERIMENT DETAILS

Platform and task definitions. We use a 6-DoF AgileX Piper arm with an external RGB camera and a wrist-mounted RGB camera. The four tasks follow the naming and order in the main text. **Place Pen in Cup (T1)** requires placing the pen in the target cup. **Place Carrot in Bowl (T2)** requires picking up the carrot and placing it in the bowl. **Place Blocks in Drawer and Close It (T3)** requires placing the blocks in the second drawer and then closing it. **Stack Three Bowls (T4)** requires forming a complete stack of three bowls.

Dataset and observation format. The dataset contains 125 demonstrations and 71,702 recorded frames collected through master–slave teleoperation. We train a single language-conditioned policy jointly on demonstrations from all four tasks.

The external and wrist cameras provide RGB images at 480×640 resolution. Each view is resized to 224×224 and the two views are concatenated horizontally to form a 224×448 video input. Both recorded proprioceptive states and actions are seven-dimensional. The Piper configuration preserves the recorded absolute action representation without converting it to a state-relative delta. State and action normalization uses the dataset’s 1st and 99th percentile statistics. A training window contains 32 actions; its video sampling and temporal compression follow Appendix A. Instruction embeddings are cached with a maximum context length of 128 tokens.

The policy uses the default one-block action expert and 16 dynamics registers described in Appendix A. Online LaWM supervision uses only the external-camera image pair; the teacher does not receive robot states or actions.

Evaluation protocol. The ID evaluation protocol specifies 10 trials per task, or 40 trials per method. Object placements are randomized across trials and matched across methods, and success rates are reported separately for each task according to the task goals above. At inference, the model uses the current two-camera observation and proprioceptive state, performs one video prefill with pure-noise future slots, and generates a 32-action chunk using 10 action-denoising steps. Demonstrated future observations and the LaWM teacher are not used during policy execution.

D DETAILED COMPONENT ABLATION

Table 9 provides the complete seven-category results for the component ablation in Section 3.1. All configurations use a 30-layer Wan2.1 Video DiT and a 30-layer Action DiT without additional policy-level embodied pretraining. The results match Table 1.

Table 9: Complementarity of Future-KV and latent-action supervision. LIBERO-Plus success rate (%), without embodied pretraining, using a 30-layer Action DiT. The baseline is Fast-WAM re-trained with Wan2.1 under matched settings. ✓/× indicate inclusion/omission. Bold marks the best result in each column.

Latent action	Future feature	Camera	Robot	Language	Light	Background	Noise	Layout	Avg
×	×	37.0	36.5	64.1	81.5	49.3	50.1	63.7	53.6
✓	×	49.0	36.8	65.1	75.9	48.5	50.8	61.7	54.8
×	✓	37.8	41.0	63.9	83.4	65.7	61.5	65.6	58.5
✓	✓	64.6	61.6	86.6	95.3	74.1	77.7	81.7	76.7

E DETAILED LIBERO RESULTS

Table 10 reports the suite-level results corresponding to the average success rates in Table 4.

Table 10: Detailed success rates (%) on the four LIBERO suites.

Method	P.T.	Spatial	Object	Goal	Long	Avg.
OpenVLA (Kim et al., 2024)	✓	84.7	88.4	79.2	53.7	76.5
WorldVLA (Cen et al., 2025b)	✓	87.6	96.2	83.4	60.0	81.8
π_0 -Fast (Pertsch et al., 2025)	✓	96.4	96.8	88.6	60.2	85.5
π_0 (Black et al., 2024)	✓	96.8	98.8	95.8	85.2	94.1
$\pi_{0.5}$ (Intelligence et al., 2025)	✓	98.8	98.2	98.0	92.4	96.9
UniVLA (Bu et al., 2025)	✓	96.5	96.8	95.6	92.0	95.2
CogVLA (Li et al., 2026b)	✓	<u>98.6</u>	98.8	96.6	<u>95.4</u>	97.4
Fast-WAM (Yuan et al., 2026b)	×	98.2	100.0	<u>97.0</u>	95.2	<u>97.6</u>
ForeWAM	×	97.6	<u>99.8</u>	98.0	96.2	97.9

F BENCHMARK EVALUATION PROTOCOLS

LIBERO and LIBERO-Plus. We evaluate our model on LIBERO (Liu et al., 2023) and LIBERO-Plus (Fei et al., 2025). For LIBERO, we follow the standard benchmarking protocol and train on the four standard suites: Spatial, Object, Goal, and LIBERO-Long. Each suite contains 500 expert demonstrations across 10 tasks.

LIBERO-Plus extends the LIBERO tasks with additional visual and layout variations. For training, we use the original LIBERO demonstrations and exclude the augmented LIBERO-Plus training data. We evaluate the trained policies under the LIBERO-Plus protocol and report overall success rate. For every table reporting Avg, we aggregate evaluation episodes as

$$\text{Avg} = \text{Overall SR} = 100 \times \frac{\sum_c S_c}{\sum_c N_c}, \quad (9)$$

where S_c and N_c are the numbers of successful and total evaluation episodes in group c , respectively (e.g., a LIBERO suite or a LIBERO-Plus perturbation category). Thus, Avg weights each group by its episode count; it is not the unweighted arithmetic mean of category percentages unless all groups contain the same number of episodes. Aggregation uses the underlying counts before rounding.

RoboCasa. We evaluate on 24 static kitchen manipulation tasks with a single Franka Emika Panda arm. Each task uses five scenes with different floor plans and visual styles, with 10 trials per scene. We report the mean success rate over all 24 tasks and three random seeds, for 3,600 trials in total. The evaluation uses unseen object instances, and two of the five scenes for each task contain styles absent from training.

We train ForeWAM on the successful human demonstrations in the released RoboCasa dataset and do not use the additional MimicGen demonstrations.

Real-world Experiments. Our real-world setup consists of a 6-DoF AgileX Piper arm with third-person and wrist-mounted RGB cameras. The task suite includes four tabletop tasks: **Place Pen**

in Cup (T1), Place Carrot in Bowl (T2), Place Blocks in Drawer and Close It (T3), and Stack Three Bowls (T4). These tasks cover precision placement, pick-and-place, articulated-object manipulation, and multi-stage stacking. In T3, the target is the second drawer. The training dataset contains 125 robot demonstrations collected through master–slave teleoperation, totaling 71,702 frames across the four tasks. We train ForeWAM jointly on demonstrations from all four tasks. The ID evaluation protocol specifies 10 trials per task, with object placements randomized across trials and matched across methods. We also evaluate out-of-distribution (OOD) robustness by perturbing object attributes in pen placement and drawer manipulation (Figure 4).